

A Principle of Sherlock Holmes and its uses in Mathematics

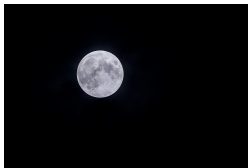
Martin Liebeck

Imperial College London

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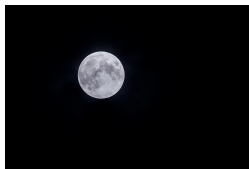
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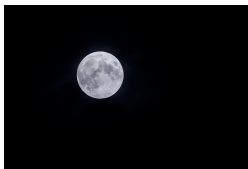
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But sometimes you can prove things exist without exhibiting them.

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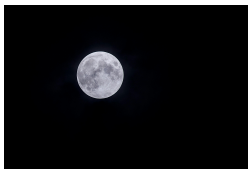


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Eg. There are two people in Perth with the same number of hairs on their head.

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My theme Proving things in maths exist without exhibiting them.
There are many methods for doing this, but I'll focus on what I call the **Sherlock Holmes method**.

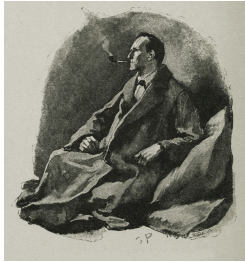


Figure: Sherlock Holmes by Sidney Paget

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- You want to prove a certain object or structure exists.
- Eliminate all “universes” in which the object does not exist.
- Show that some possible universes remain. In such universes, the object exists! QED

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The argument is often phrased in the language of probability: estimate the probability P of all possible universes in which the structure does not exist, i.e.

$$P = \frac{\text{no. of bad universes}}{\text{total no. of possible universes}}.$$

Show that $P < 1$. Conclusion: structure exists!

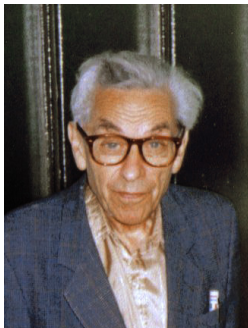


Figure: Paul Erdős by Kmhmh

Probabilistic method was pioneered by the great Hungarian mathematician Paul Erdős (1913-1996).

Has been used in number theory, combinatorics, algebra, computer science,.....

Ramsey theory: order in chaos

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Figure: Frank Ramsey, Cambridge Wittgenstein archive

Frank Ramsey (1903-1930)

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Ramsey's Friends/Strangers theorem: In any party of 6 people, either

- there are 3 people all of whom know each other, or
- there are 3 people who are strangers to each other.

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- there are 3 people all of whom know each other, or
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To think about this, we convert such a party into a **graph**:

each person is represented by a point,

join two people by a red edge if they know each other,

join two people by a blue edge if they don't know each other.

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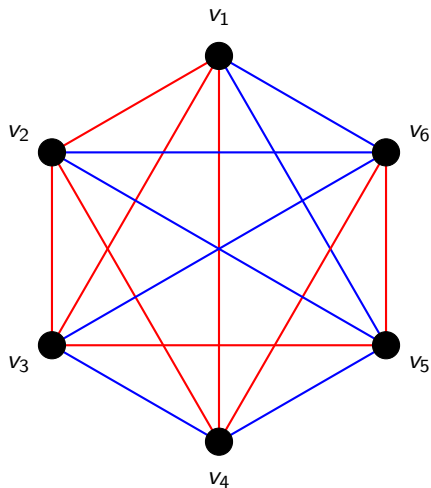
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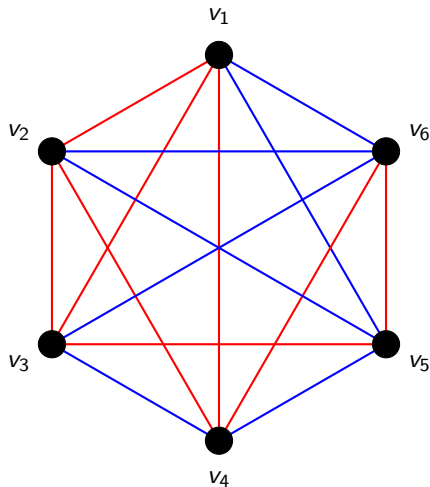
join two people by a blue edge if they don't know each other.

Here's an example of such a coloured graph:

Example: A Red/Blue Colouring of K_6



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Ramsey's theorem says: **any** red/blue coloured K_6 must have either a **red triangle** or a **blue triangle**.

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Bad idea: there are 2^{15} possible colourings. Check each of them has a red or blue triangle.

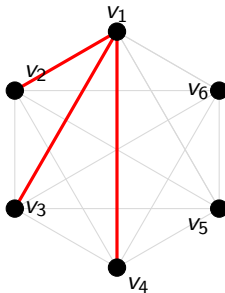
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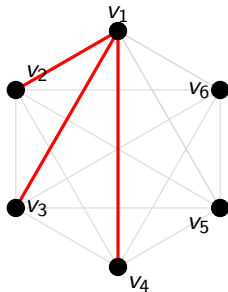
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Ramsey's idea: Consider the point v_1 . It has **5 edges** to the other 5 points. At least 3 of these edges share the same colour.

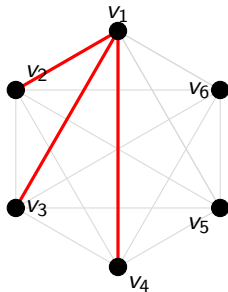
Let's assume there are 3 red edges from v_1 :



Ramsey's theorem

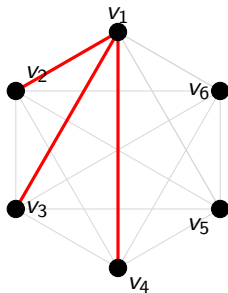


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Now consider the 3 edges among v_2, v_3, v_4 .

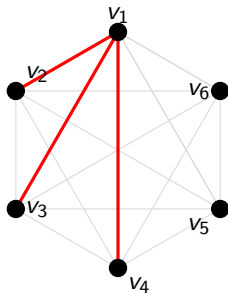
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Now consider the 3 edges among v_2, v_3, v_4 .

If they are all blue, there is a **blue triangle** v_2, v_3, v_4 .

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Now consider the 3 edges among v_2, v_3, v_4 .

If they are all blue, there is a **blue triangle** v_2, v_3, v_4 .

On the other hand, if one of them is red, say the $v_2 v_3$ edge, then there is a **red triangle** v_1, v_2, v_3 . QED!!

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Graph version: Any red/blue coloured K_6 must have either a **red triangle** or a **blue triangle**.

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Can we generalise – is there a number N , such that in any party of N people, either

there are r all of whom know each other, or
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The smallest N is called the **Ramsey number** $R(r, s)$.

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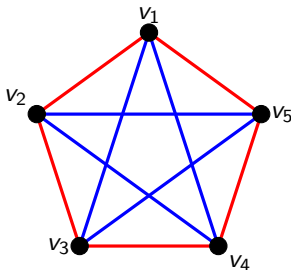
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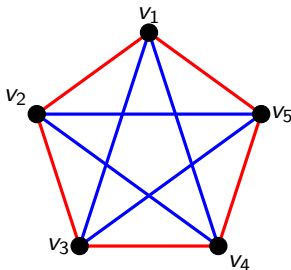
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So $R(3, 3) = 6$.

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Erdős: “Imagine an alien force, vastly more powerful than us, landing on Earth and demanding the value of $R(5, 5)$ or they will destroy our planet. In that case, we should marshal all our computers and all our mathematicians and attempt to find the value. But suppose, instead, that they ask for $R(6, 6)$. In that case, we should attempt to destroy the aliens.”

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Nevertheless, using the Sherlock Holmes method, Erdős proved

Theorem (1947) For any r , $R(r, r) > 2^{r/2}$.

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Ramsey number $R(r, r)$ is the smallest N such that any red/blue colouring of K_N must have a K_r which is completely red or completely blue.

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The total number of sets of 20 points is the “binomial coefficient” $\binom{1024}{20}$. So

$$P \leq \binom{1024}{20} \times \frac{2}{2^{190}} \approx 0.000000000000000085.$$

Conclusion: there exists a colouring of K_{1024} with no red or blue K_{20} . In fact the vast majority of colourings have no red or blue K_{20} .

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Order in chaos: for $N = R(r, r)$, a random red/blue colouring of K_N (chaos) must have a K_r that is completely red or completely blue (order).

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There is no 3-term arithmetic progression (AP) that is all red or all blue.

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van der Waerden's Theorem (1927) There is a number $N = W(k)$ such that for any red/blue colouring of the numbers $1, \dots, N$, there is always a red or blue AP of length k .

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Known: $W(3) = 9$, $W(4) = 35$, $W(5) = 178$, $W(6) = 1132$, $W(7) = ?$

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Infinite version: colour the natural numbers $1, 2, 3, \dots$ with a finite number of colours $C_1, \dots, C_r$. Then one colour has AP's of arbitrary length.

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Ultimate version: the prime numbers

2, 3, 5, 7, 11, ..., 1009, 1013, 1019, 1021, 1031, ...,
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The list seems a bit chaotic. Is there some order? Are there long AP's?

Length 4: 5, 11, 17, 23

Length 5: 5, 11, 17, 23, 29

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Ramsey theory

van der Waerden's Theorem There is a number $N = W(k)$ such that for any red/blue colouring of the numbers $1, \dots, N$, there is always a red or blue AP of length k .

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- if A and B are matrices in G , their product AB is also in G
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To get a **finite** group, we take the numbers a, b, c, d in a **finite field** $\mathbb{F}_p$, where p is prime. $\mathbb{F}_p$ is the numbers $0, 1, 2, \dots, p-1$, where we add and multiply x, y by taking the remainder on dividing $x+y$ and xy by p .

Eg. In $\mathbb{F}_5$, $3+4=2$ and $3 \times 4=2$.

Group theory

So our group is the set of 2×2 matrices

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$\langle A, B \rangle$ is a **subgroup** of $\text{SL}(2, p)$.

Question Do there exist matrices A, B such that $\langle A, B \rangle = \text{SL}(2, p)$?

If so, can just use A and B to input the whole of $\text{SL}(2, p)$.

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Special linear group $\mathrm{SL}(2, p)$: the group of all matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a, b, c, d are in $\mathbb{F}_p$ and $ad - bc = 1$.

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Using this, we can give a Sherlock-style proof of

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If $\langle A, B \rangle \neq G$ then A, B are in one of the subgroups M in (1)-(3). For such M ,

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Figure: Cheryl Praeger by John Henstridge

Thank you Cheryl!